# DISPATCH DECISION TRANSPARENCY & AUDIT TRAIL

A proposed explainability and record-keeping design for emergency allocation

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**Project:** Ambulance and Emergency Dispatch System

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**Purpose:** Make dispatch outcomes easier to understand, review and debug.

## 1. The unique idea: explain every dispatch decision

Instead of showing only which ambulance was assigned, the system can also show the main reasons behind the choice: emergency priority, ambulance availability, estimated route distance, and hospital bed availability. This makes the result easier for a reviewer or operator to understand.

## 2. Suggested decision record

| Field | What to record | Why it helps |
|---|---|---|
| Request ID | Unique identifier for the emergency request. | Links the decision to one request. |
| Priority | Priority level and severity category. | Shows why urgent requests are handled first. |
| Ambulance choice | Selected ambulance and its availability at assignment time. | Helps verify eligible selection. |
| Route estimate | Calculated route distance or route status. | Makes route comparison reviewable. |
| Hospital decision | Selected hospital and available-bed count used. | Explains hospital selection. |
| Outcome | Assigned, rejected, no resource, or validation error. | Supports debugging and test review. |

## 3. Example of an explainable result

**Illustrative example only:** “Critical request received. Ambulance 2 was selected because it was available and eligible, with a calculated route distance of 5 km. City Hospital was considered with 5 available beds. The assignment was recorded successfully.” Replace these sample values with actual output from your program.

## 4. Suggested audit flow

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|---|---|---|---|
| **1. Receive**<br>Emergency request | **2. Evaluate**<br>Priority + availability | **3. Select**<br>Ambulance + route | **4. Record**<br>Choice + outcome |

## 5. Benefits

- **Transparency:** the reason for a selection is visible.
- **Debugging:** failed or unexpected assignments are easier to trace.
- **Testing:** recorded decisions can be compared against expected results.
- **Accountability:** the team can review which inputs influenced a dispatch outcome.